CS2580

UNIVERSITY OF WARWICK

Second Year Winter Examinations 2013/14

Database Systems

Time allowed: 1.5 hours

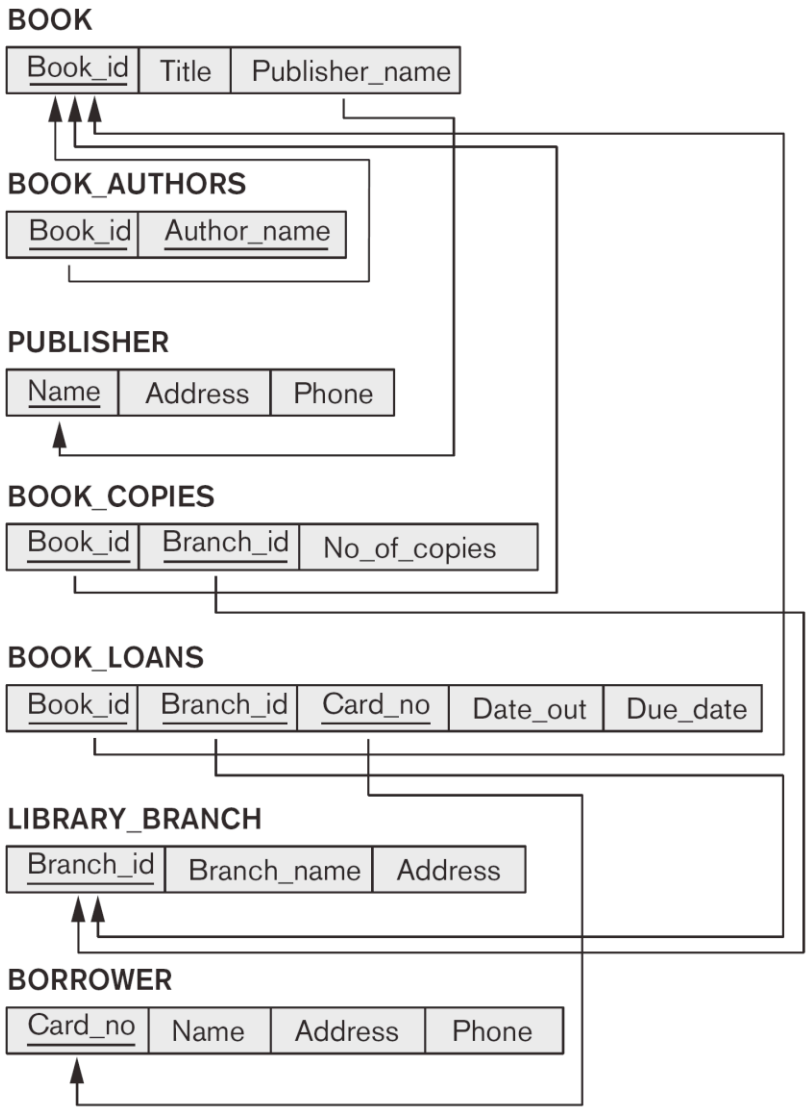
Answer **FOUR** questions out of six.

Read carefully the instructions on the answer book and make sure that the particulars required are entered on each answer book.

Calculators are not required and not permitted.

(continued)

1. Consider the LIBRARY relational database schema depicted below. Choose an appropriate action (reject, cascade, set to null, set to default) for each referential integrity constraint, both for the *deletion* of a referenced tuple, and for the *update* of a primary key attribute value in a referenced tuple. Justify your choices. [25]



(continued)

2. Consider a database for hotel bookings:

   HOTEL (hotelNo, hotelName, city)
   ROOM (roomNo, hotelNo, type, price)
   BOOKING (hotelNo, guestNo, dateFrom, dateTo, roomNo)
   GUEST (guestNo, guestName, guestAddress)

   Specify the following queries in SQL:

   (a) List the numbers and names of all hotels in London. [5]
   (b) List the names and addresses of all guests living in London, alphabetically ordered by name. [5]
   (c) List the bookings for which no dateTo has been specified. [5]
   (d) List the details of all rooms at the Grosvenor Hotel, including the name of the guest staying in the room, if the room is occupied. [5]
   (e) What is the total income from bookings for the Grosvenor Hotel today? [5]

(continued)

3. Specify the following views in SQL on the COMPANY database schema depicted below.

   (a) A view that has the department name, manager name, and manager salary for every department. [5]
   (b) A view that has the employee name, supervisor name, and employee salary for each employee who works in the 'Research' department. [6]
   (c) A view that has the project name, controlling department name, number of employees, and total hours worked per week on the project for each project. [7]
   (d) A view that has the project name, controlling department name, number of employees, and total hours worked per week on the project for each project with more than one employee working on it. [7]

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

Essn	Pno	Hours

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship

(continued)

4. Express in relational algebra the following queries on the COMPANY database schema depicted below.

   (a) Find the names of employees that are directly supervised by 'Franklin Wong'. [5]
   (b) Retrieve the names of all employees in department 5 who work more than 10 hours per week on the 'ProductX' project. [6]
   (c) Retrieve the names of employees who work on every project. [7]
   (d) For each department, retrieve the department name, and the average salary of employees working in that department. [7]

EMPLOYEE

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno

DEPARTMENT

Dname	Dnumber	Mgr_ssn	Mgr_start_date

DEPT_LOCATIONS

Dnumber	Dlocation

PROJECT

Pname	Pnumber	Plocation	Dnum

WORKS_ON

Essn	Pno	Hours

DEPENDENT

Essn	Dependent_name	Sex	Bdate	Relationship

(continued)

5. Consider the following relation for published books:

   BOOK (Book_title, Author_name, Book_type, Listprice, Author_affil, Publisher)

   Author_affil refers to the affiliation of the author. Suppose the following dependencies exist:

   Book_title -> Publisher, Book_type
   Book_type -> Listprice
   Author_name -> Author_affil

   (a) What normal form is the relation in? Explain your answer. [10]

   (b) Apply normalization until you cannot decompose the relations further. State the reasons behind each decomposition. [15]

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6. What are the typical security classifications? Outline the Bell-LaPadula model, and discuss the simple security property and the $*$-property, explaining the justification behind these rules for enforcing multilevel security. [25]

(End)